

Politecnico di Torino
Department of Electronics and Telecommunications
Master's degree in Mechatronic Engineering



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Sensors, Embedded Systems and Algorithms for Service Robotics

Report 02

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Group n° 103

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1 OBJECTIVE

The aim of this laboratory activity is to

- Develop a ROS 2 Package to run EKF on the robot
- Localize your robot using landmark measurements.
- Fuse measurements from different sensors (odometry,IMU) to improve localization accuracy.

2 DESIGN

The objective of this work is to design and implement an Extended Kalman Filter (EKF) for robot localization using probabilistic motion and measurement models, and to validate it in both simulation and real-world scenarios. The design process is structured progressively across the tasks, starting from model definition and culminating in deployment on a real robot.

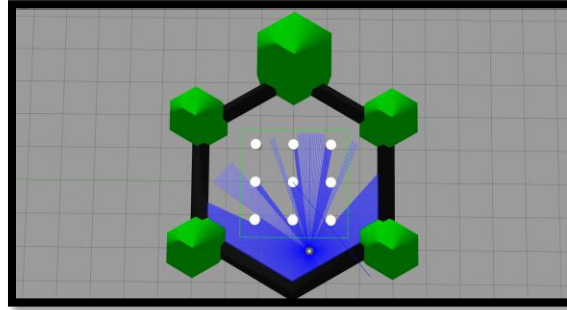


Figure 1 – Burger Simulation in Gazebo

The robot state is defined as $x = [x, y, \theta]^T$, representing its position and orientation in the global reference frame. Control inputs are the linear and angular velocities $u = [v, \omega]^T$. Robot motion is modeled using the nonlinear velocity-based motion model, which accurately represents the kinematics of a differential-drive robot through trigonometric relations. Motion uncertainty is incorporated by modeling noise on the effective linear and angular velocities, as well as an additional orientation drift term, resulting in a probabilistic motion model.

Since the motion model is nonlinear, it is linearized to be compatible with the EKF framework. This is achieved by computing the Jacobian matrices with respect to the state and the control inputs, denoted as $G = \frac{\partial g(x,u)}{\partial x}$ and $V = \frac{\partial g(x,u)}{\partial u}$. These Jacobians are computed symbolically using SymPy to obtain exact analytical expressions and ensure numerically stable covariance propagation..

A landmark-based measurement model is used to correct the predicted state. Landmarks are assumed to have known positions in the global frame, while the robot measures their range and bearing. The nonlinear measurement function maps the robot state to expected sensor observations and is linearized using the Jacobian $H = \frac{\partial h(x)}{\partial x}$, which captures how pose uncertainty affects the predicted measurements.

Building on these models, the EKF is implemented as a ROS 2 node. The prediction step runs at a fixed rate of 20 Hz using velocity data from the `/odom` topic, while the update step is triggered asynchronously whenever landmark

measurements are received. The EKF state estimate is published as a *nav_msgs/Odometry* message on the */ekf* topic, enabling direct comparison with odometry and ground truth data in simulation.

Finally, the same EKF design is deployed on the real TurtleBot3 robot without modifying the filter structure. Landmark measurements from the onboard camera are processed through the same measurement model, while odometry is used for prediction. Although ground truth is unavailable on the real robot, qualitative comparison between EKF and odometry trajectories demonstrates the robustness and portability of the designed localization system.

3 EXPERIMENTS

3.1 SIMULATION

3.1.1 TASK 0

The velocity-based probabilistic motion model was implemented using the structure provided in the *probabilistic-robotics-python-examples* package. The program includes functions for sampling Gaussian noise, applying the nonlinear motion model, and evaluating the resulting distributions. Starting from an initial robot state and fixed control commands, multiple pose samples were generated to visualize the dispersion induced by motion uncertainty. The procedure was also applied across a sequence of commands to show how the uncertainty accumulates with each movement step.

To study the influence of different uncertainty sources, two distinct sets of noise parameters were tested.

First Configuration : $a = [0.08, 0.06, 0.001, 0.002, 0.05, 0.02]$

Second Configuration: $a = [0.001, 0.002, 0.08, 0.07, 0.05, 0.02]$

The first configuration introduced higher variability in the linear motion, while the second emphasized angular uncertainty. For each case, the sampled poses were plotted in the (x, y) plane, allowing a direct comparison of how the distribution spreads under different noise characteristics.

For the measurement model (Point B), the landmark-based probabilistic sensor model was implemented using the structure provided in the *probabilistic-robotics-python-examples* package. This program includes functions like *landmark_range_bearing_sensor()* and *landmark_model_sample_pose()* to generate samples of the measured range and bearing using an initial robot pose and fixed landmark location. A set of noisy measurements was sampled according to the specified standard deviations, and the corresponding pose samples were plotted to visualize the resulting distribution in the (x, y) space. This experiment illustrates how measurement noise affects the inferred robot position when only landmark observations are considered.

3.1.2 TASK 1

```
sec: 297
nanosec: 850000000
frame_id: odom
child_frame_id: base_footprint
pose:
  pose:
    position:
      x: -1.999924630152992
      y: -0.500001023713533
      z: 0.008532241263603206
    orientation:
      x: 4.934785687654057e-05
      y: 0.002869129104513429
      z: 3.0290011602840686e-06
      w: 0.9999958828152912
  covariance:
    - 1.0e-05
    - 0.0
```

Figure 2 – Data sample from */odom*

```
- id: 11
  hamming: 0
  goodness: 1.0
  decision_margin: 1.0
  range: 1.07772958278656
  bearing: -0.5887922048568726
- id: 12
  hamming: 0
  goodness: 1.0
  decision_margin: 1.0
  range: 1.0231660604476929
  bearing: 0.4875313341617584
- id: 21
  hamming: 0
  goodness: 1.0
  decision_margin: 1.0
  range: 2.078582525253296
  bearing: -0.3059083819389343
```

Figure 3 – Data sample from */Landmarks*

The EKF localization algorithm was evaluated in a simulated TurtleBot3 environment using Gazebo. The simulation was launched with a predefined landmark configuration, and the robot was manually driven using keyboard teleoperation to generate motion and landmark observations.

```
nanosec: 822249330
frame_id: map
child_frame_id: ''
pose:
  pose:
    position:
      x: -1.9972584839416845
      y: -0.41723411732077004
      z: 0.0
    orientation:
      x: 0.0
      y: 0.0
      z: -0.025949081350297828
      w: 0.9996632658936087
  covariance:
    - 0.0
    - 0.0
    - 0.0
```

Figure 4 – Data sample from */ekf*

During the experiment, the EKF node ran continuously, performing state prediction at a fixed rate of 20 Hz using velocity data from the */odom* topic and updating the state whenever landmark measurements were received on the */landmarks* topic. The estimated robot pose was published on the */ekf* topic as a *nav_msgs/Odometry* message.

To evaluate localization performance, the */ground_truth*, */odom*, and */ekf* topics were recorded using rosbag while the robot executed a sequence of translational and rotational motions.

ros2 bag record /ground_truth /odom /ekf

The recorded data were processed offline to extract the robot position and orientation over time. The extracted signals were then plotted using **PLOTJUGGLER** to compare the estimated pose provided by the EKF with raw odometry and ground truth data, and to analyze the resulting trajectories in the planar workspace.

3.1.3 TASK 2

For the sensor fusion task, the system state used in Task 1, $[x, y, \theta]^T$, was extended to $[x, y, \theta, v, \omega]^T$ in order to incorporate measurements from the IMU and wheel encoders. This extension allows linear and angular velocities to be explicitly estimated as part of the EKF state. The prediction step is based on the same motion model used in Task

1, with the difference that the velocities v and ω are now taken directly from the state vector rather than being read from the odometry topic. During prediction, these velocity components are assumed to remain constant and are subsequently corrected during the update phase. As a result of the extended state, the covariance matrix is expanded to a 5×5 matrix, with the upper 3×3 block corresponding to the pose uncertainty as in the original filter. The correction step is performed using separate update functions for each sensor source (landmarks, odometry, and IMU). In each case, the Kalman gain is computed using the appropriate measurement Jacobian, the predicted covariance, and a measurement noise covariance matrix R whose size depends on the sensor. The IMU update uses a scalar R for angular velocity measurements, while odometry and landmark updates use a 2×2 matrix. The values of R reflect the assumed confidence in each sensor and are used to compute the corrected state and covariance estimates..

3.2 REAL ROBOT

For Task 3, we chose to use the EKF implementation developed in Task 1, which does not include any advanced sensor-fusion techniques. The experiment was repeated three times in order to obtain a larger dataset and support more reliable conclusions. In all runs, the robot was initially placed in the same nominal position and then manually driven via teleop within the provided map. The goal was to make the robot follow similar trajectories across all experiments, explore the full set of landmarks, and finally return to the starting point. All movements were performed at moderate speed to reduce wheel slippage and improve the quality of the odometry data. The only difference with respect to the EKF implementation used in simulation concerns the modeling of the landmarks. In the real robot setup, landmarks are characterized by a three-dimensional position, whereas in Gazebo they are assumed to lie on the ground plane. As a consequence, the computation of the range measurement also includes the contribution along the z-axis. In particular, the height of the camera mounted on the robot is estimated to be approximately 0.2 m, and this value is taken into account when computing the Euclidean distance between the robot and each landmark.

4 RESULTS AND DISCUSSION

4.1 Task 0

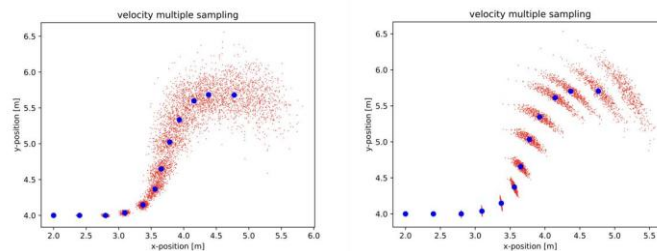


Figure 5 – Noise Affecting linear velocity Figure 6 – Noise Affecting Angular Velocity

When the noise parameters primarily affected the linear velocity, the sampled trajectories showed a gradual widening along the direction of motion. The dispersion increased progressively at each step, but the overall shape of the trajectory remained consistent with the intended path. In contrast, when angular noise was dominant, the sampled poses exhibited a much stronger divergence. Small orientation errors introduced at each step accumulated over time, causing the trajectories to fan out along curved paths. This resulted in crescent-shaped distributions in the (x,y) plane, demonstrating that angular uncertainty has a more significant impact on long-term localization accuracy than linear uncertainty.

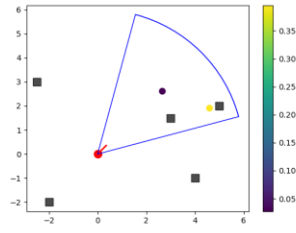


Figure 7 – Robot Range using Landmark Model

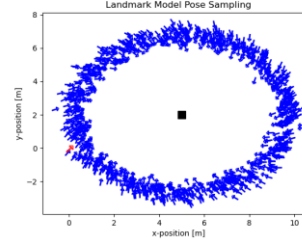


Figure 8 – Landmark Pose by samples of Robot pose

For the landmark-based measurement model, sampling from a fixed range and bearing measurement produced a spatial distribution of possible robot poses around the landmark. The resulting pose samples formed characteristic geometric patterns consistent with the polar nature of the measurement model. Range noise caused radial dispersion around the landmark, while bearing noise resulted in angular spreading of the sampled poses. The combined effect of these uncertainties produced a broadened region of likely robot positions, centered around the true pose. Overall, the experimental results confirm that both motion and measurement uncertainties lead to distinct and predictable distributions in the robot pose space. The motion model experiments illustrate how uncertainty propagates through the robot kinematics over time, while the measurement model results demonstrate how sensor noise affects pose inference from landmark observations.

Finally, both the motion and measurement models were linearized by computing their analytical Jacobians using SymPy. The symbolic computation ensured exact expressions for the motion Jacobians G and V , and for the measurement Jacobian H , which will later be required for the prediction and correction steps of the Extended Kalman Filter.

$$G = \frac{\partial g}{\partial x} = \begin{bmatrix} 1 & 0 & \frac{v}{w}(\cos(\theta + w\Delta t) - \cos\theta) \\ 0 & 1 & \frac{v}{w}(\sin(\theta + w\Delta t) - \sin\theta) \\ 0 & 0 & 1 \end{bmatrix}$$

$$v = \frac{\partial g}{\partial u} = \begin{bmatrix} \frac{1}{w}(\sin(\theta + w\Delta t) - \sin\theta) & \frac{v}{w^2}(\sin\theta - \sin(\theta + w\Delta t)) + \frac{v}{w}\Delta t \cos(\theta + w\Delta t) \\ \frac{1}{w}(\cos\theta - \cos(\theta + w\Delta t)) & \frac{v}{w^2}(\cos(\theta + w\Delta t) - \cos\theta) + \frac{v}{w}\Delta t \sin(\theta + w\Delta t) \\ 0 & \Delta t \end{bmatrix}$$

$$H = \begin{bmatrix} \frac{-mx + x}{\sqrt{(mx - x)^2 + (my - y)^2}} & \frac{-my + y}{\sqrt{(mx - x)^2 + (my - y)^2}} & 0 \\ \frac{my - y}{(mx - x)^2 + (my - y)^2} & \frac{-mx + x}{(mx - x)^2 + (my - y)^2} & -1 \end{bmatrix}$$

4.2 Task 1

The plots compare the robot pose estimated by the EKF with raw odometry and ground truth data obtained from the simulation. The results are reported for the three state components x , y , and θ , as well as for the planar trajectory in the (x, y) space.

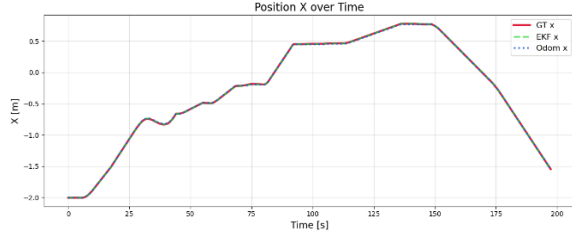


Figure 9 - Plot of state X over time

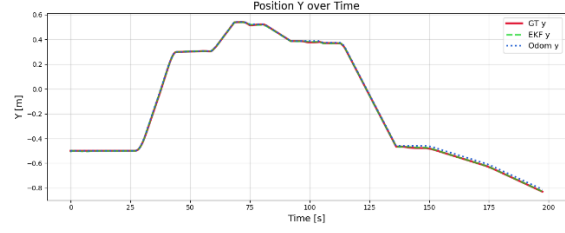


Figure 10 - Plot of state Y over time

From the position plots along x and y axis, it can be observed that the EKF as well as Odometry estimate closely follows the ground truth throughout the entire experiment. Which tells that there are no highly differential error while implementing both EKF and Odometry for the robot in the simulation world. Although, small deviation can be seen during longer motion segments while plotting state y over time with respect to Odometry which are caused by accumulation of motion errors.

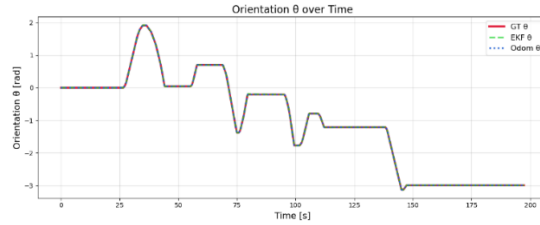


Figure 11 - Plot of state θ over time

The orientation plot shows a similar behaviour. While the odometry orientation exhibits slight discrepancies during rotational maneuvers, the EKF estimate aligns well with ground truth, particularly during sharp heading changes.

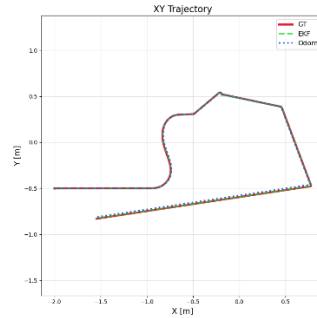


Figure 12 - Trajectory of Robot in Simulation

The odometry trajectory deviates from the true path due to the cumulative integration errors over a long run, whereas the EKF trajectory remains tightly aligned with the ground truth over the entire motion.

State	RMSE with respect to Ground Truth	MAE with respect to Ground Truth
X	0.0028m	0.0021m
Y	0.0025m	0.0019m
XY	0.0037m	0.0031m
Orientation θ	0.0019 rad	0.0013 rad

Table 1 - Metrics of EKF with respect to Ground Truth

State	RMSE with respect to Ground Truth	MAE with respect to Ground Truth
X	0.0049m	0.0036m
Y	0.0103m	0.0073m
XY	0.0114m	0.0091m
Orientation θ	0.0091 rad	0.0029 rad

Table 2 – Metrics of Odometry with respect to Ground Truth

The quantitative evaluation confirms the improvement achieved by the EKF over raw odometry. With respect to ground truth, the EKF achieves significantly lower errors in both position and orientation. The planar position is reduced from 0.0114m (Odometry) to 0.0037m (EKF), while the orientation RMSE decreases from 0.0091 rad to 0.0019rad. A similar improvement observed in MAE metrics, where the EKF consistently shows smaller errors than odometry across all state components.

Overall, the EKF filter designed by us provide a more accurate and stable localization estimate than raw odometry alone.

4.2 Task 2

The results show a strong agreement between the Extended Kalman Filter with sensor fusion and the ground truth (GT) across all evaluated variables, and an improvement with respect to pure odometry.

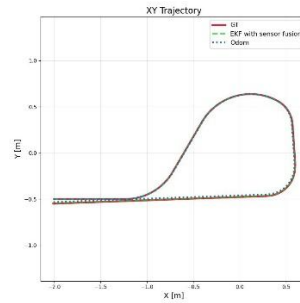


Figure 13 – Trajectory of Robot in Simulation

The planar trajectory comparison highlights that the EKF estimate closely follows the ground truth along the entire path. The odometry-only trajectory shows small but noticeable deviations, in particular when there are change of direction and tight turns, which are typical effects of wheel slip and drift. The sensor fusion approach significantly mitigates these errors by exploiting the complementary information provided by the IMU and wheels encoders.

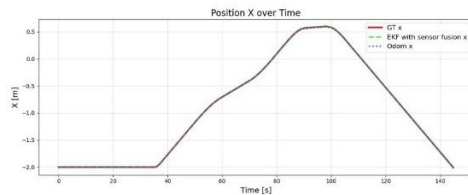


Figure 14 – Plot of state X over time

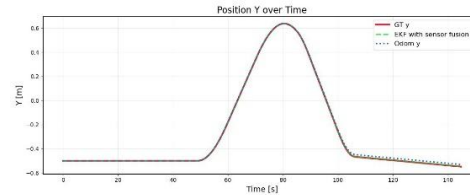


Figure 15 – Plot of state Y over time

For both $x(t)$ and $y(t)$, the EKF output almost overlaps with the ground truth signal. During dynamic phases (acceleration, turns, and trajectory changes), odometry shows a slight lag and accumulated error, especially after long intervals. The EKF maintains higher accuracy and stability, confirming that the fusion-based correction effectively compensates for odometric drift.

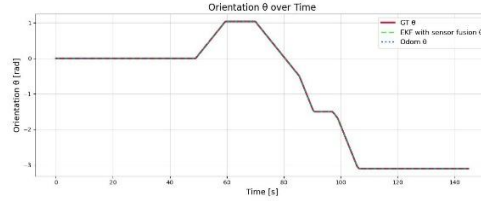


Figure 16 - Plot of state θ over time

Also the orientation estimate obtained through EKF sensor fusion closely matches the ground truth throughout the entire experiment. The IMU angular velocity measurement plays a key role in stabilizing the heading estimation, especially during rotations, where odometry alone tends to accumulate angular errors. As a result, the EKF orientation estimate remains accurate even during rapid changes.

State	RMSE with respect to Ground Truth	MAE with respect to Ground Truth
X	0.0033m	0.0024m
Y	0.0023m	0.0017m
XY	0.0040m	0.0032m
Orientation θ	0.0015 rad	0.0011 rad

Table 3 - Metrics of EKF Sensor Fusion with respect to Ground Truth

State	RMSE with respect to Ground Truth	MAE with respect to Ground Truth
X	0.0040m	0.0023m
Y	0.0102m	0.0067m
XY	0.0110m	0.0080m
Orientation θ	0.0082 rad	0.0025 rad

Table 4 - Metrics of Odometry with respect to Ground Truth

The numerical results confirm the effectiveness of the Extended Kalman Filter with sensor fusion. When compared to ground truth, the EKF achieves very low position errors, with an RMSE of 0.0040 m in the XY plane and MAE below 0.0032 m. Orientation estimation is also highly accurate, with an RMSE of 0.0015 rad, indicating excellent heading tracking. In contrast, odometry-only estimation exhibits significantly higher errors, particularly along the y axis, where the RMSE increases to 0.0102 m. The planar position error rises to 0.0110 m, and the orientation RMSE reaches 0.0082 rad, reflecting the accumulation of drift over time. Overall, the metrics show that sensor fusion reduces both position and orientation errors compared to odometry alone, demonstrating the clear advantage of the EKF-based fusion approach.

4.3 Task 3

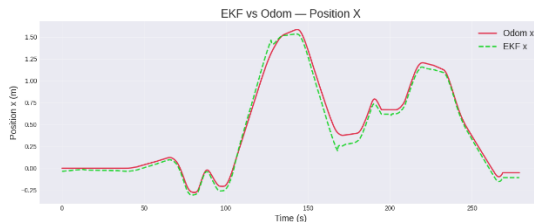


Figure 17 – State X over time w.r.t EKF & Odom

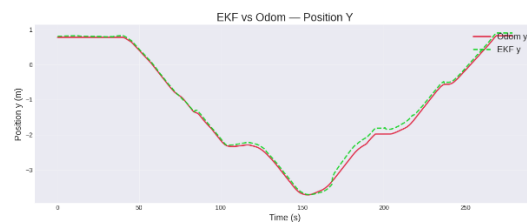


Figure 18 - State Y over time w.r.t EKF & Odom

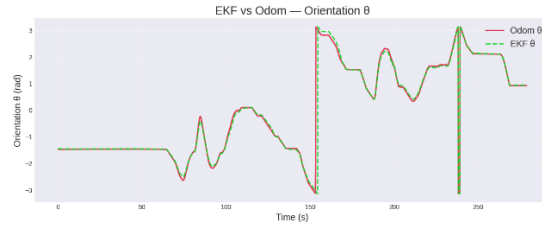


Figure 19 – Orientation θ over time w.r.t EKF & Odom

During the experiments, an initial positioning error of approximately 0.79 m was introduced, since the robot was not placed exactly at the (0,0) location defined by the map. As a result, the trajectories reported by /odom—realigned by subtracting the first recorded pose—and those estimated by the EKF, initialized in the landmark-map reference frame, exhibit a constant offset in the trajectory plots. This effect is especially visible in the x-y plane, where the two curves have nearly the same shape but start from different positions. Once this offset is compensated during visualization, the two estimates appear highly consistent throughout the entire experiment.

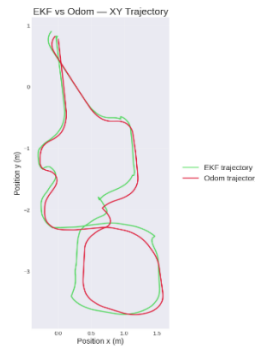


Figure 20 – Trajectory of Robot in Real World

The comparison between /odom and /ekf shows that the difference in the x and y estimates remains generally small, with larger deviations occurring mainly during tight turns and rapid direction changes. In these sections, the EKF often produces a smoother trajectory than odometry, which is more sensitive to wheel slippage and high angular velocities. This behavior is expected, as the EKF incorporates landmark measurements that can correct pose estimates in dynamically challenging situations.

Overall, the EKF trajectory appears smoother and more regular than the odometry-based one, indicating reduced noise and limited drift compared to dead reckoning. The orientation estimates from /odom and /ekf are largely consistent throughout the experiment, with apparent discontinuities caused by angle wrap-around at $\pm\pi$ rather than true heading changes. Although ground truth data were not available, the consistent behavior across multiple runs and the EKF's improved performance during sharp maneuvers confirm its effectiveness in enhancing localization accuracy over odometry alone.

The figures for the two additional runs are reported below, in both post-processed and raw formats.

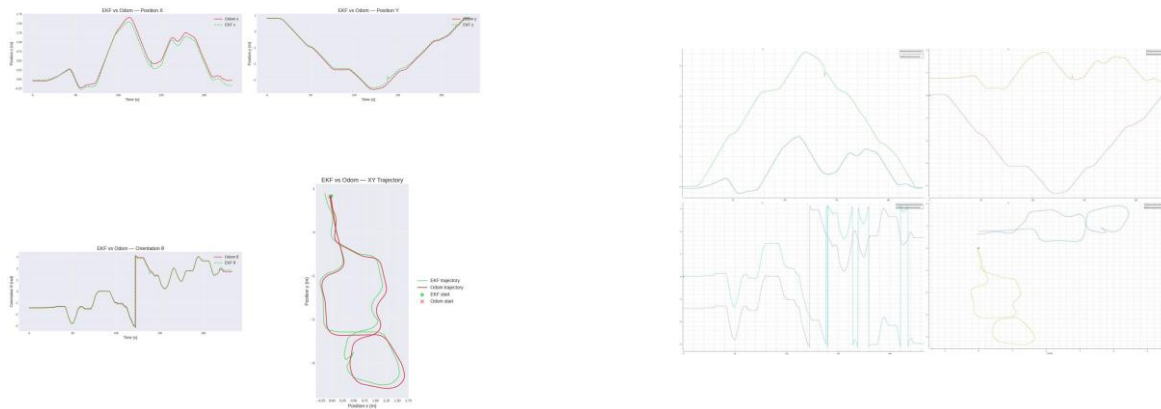


Figure 21 – Second Run of the Robot in the test Field

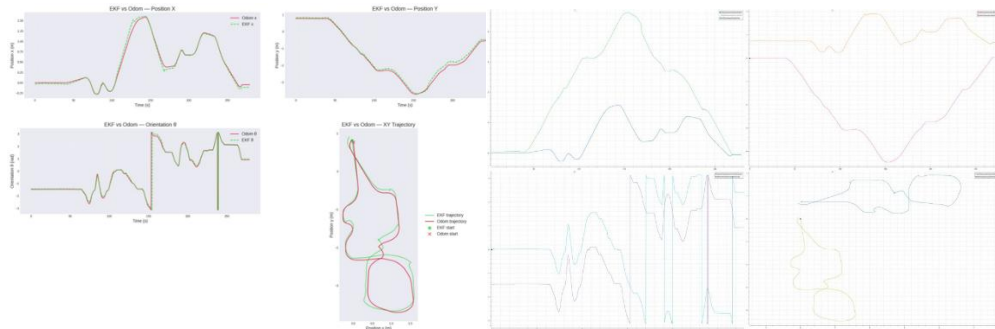


Figure 22 - Third Run of the Robot in the test Field